

Teacher Guide — senior / module-07-eleven-multiplication

IKS Curriculum

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Overview

×11 Multiplication — Ekādhikena Pūrveṇa

Band: Senior (Grades 9–12) · **Sessions:** 10 × 45 min · **Sanskrit:** Ekādhikena Pūrveṇa

Overview

A mental-arithmetic method for multiplying any number by 11 instantly: split the digits, add adjacent pairs, place sums in between. Tirthaji's *Vedic Mathematics* (1965) labelled this an application of the *Ekādhikena Pūrveṇa* sūtra. Honest history is taught alongside: this is a 20th-century mental-arithmetic system, not strictly ancient mathematics.

NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems
- §4.23 — Experiential learning (mental-math practice)
- §4.27 — Local context and language

Key terms (this module)

- *Ekādhikena Pūrveṇa* — “by one more than the previous” — sūtra 1 in Tirthaji
- *sūtra* — aphorism / formula
- *Anurūpyena* — “proportionately” — sūtra used for ×12, ×13 extensions

Prerequisites

Module 2 (Panchabhūta) is recommended before this module. Modules 5 (Bindu Paddhati / dot addition) and 7 (×11 Multiplication) are useful prerequisites for Module 9 and 11.

Materials needed (across 10 days)

- Notebook + pencils per student
- Whiteboard / chart paper
- (See `lesson-plan.md` for per-day material lists)

10-day map

- **Day 1** — The Hook — What Is the Trick?: Compute 23×11 , 35×11 mentally using the split-and-sum method.
- **Day 2** — The Carry Case: Compute 87×11 , 65×11 correctly using the carry rule.
- **Day 3** — Why It Works — The Algebra: Derive $(10a+b) \times 11 = 100a + 10(a+b) + b$ and explain.
- **Day 4** — 3-Digit $\times 11$: Compute $234 \times 11 = 2_5_7_4$ using the extended split method.
- **Day 5** — Speed Day: Complete a 50-problem mental-math sprint in 5 minutes.
- **Day 6** — Anurūpyena — $\times 12$ to $\times 19$: Extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.
- **Day 7** — Limits of the Method: Recognise when traditional long multiplication is faster than the sūtra method.
- **Day 8** — Tirthaji and Historical Context: Explain who Tirthaji was and why scholars debate the “Vedic” attribution.
- **Day 9** — Mixed Practice: Solve a mixed problem set using best-fit method per problem.
- **Day 10** — Final Assessment: Take the summative quiz and reflect on which method works best.

Files in this pack

- lesson-plan.md — detailed 10-day arc
- sources.md — sources cited in this module
- teacher-notes.md — common misconceptions, differentiation
- parent-guide.md — what parents should know
- days/day-NN-*.md — 10 per-day lesson plans
- quizzes/formative.md, quizzes/summative.md
- activities/activity-01-*.md
- assessments/rubric.md, assessments/project-brief.md
- student-workbook.md
- writing-prompts.md (or oral-recall.md for Primary)
- homework.md
- slides/deck.md

Voice

Analytical, comparative, primary-source aware; 20–30 words on average. Pitches the concept first, frames the tradition as historical context, and uses IAST for all Sanskrit terms.

10-Day Lesson Plan

Lesson Plan — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Senior)

10-day arc, 45 min per session.

Day 1 — The Hook — What Is the Trick?

Objective: Compute 23×11 , 35×11 mentally using the split-and-sum method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — The Carry Case

Objective: Compute 87×11 , 65×11 correctly using the carry rule.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Why It Works — The Algebra

Objective: Derive $(10a+b) \times 11 = 100a + 10(a+b) + b$ and explain.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — 3-Digit $\times 11$

Objective: Compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — Speed Day

Objective: Complete a 50-problem mental-math sprint in 5 minutes.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Anurūpyena — $\times 12$ to $\times 19$

Objective: Extend the method to $\times 12$ and $\times 13$ using one-more-digit adjustments.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — Limits of the Method

Objective: Recognise when traditional long multiplication is faster than the sūtra method.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Tirthaji and Historical Context

Objective: Explain who Tirthaji was and why scholars debate the “Vedic” attribution.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice

Objective: Solve a mixed problem set using best-fit method per problem.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Take the summative quiz and reflect on which method works best.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Teacher Notes

Teacher Notes — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Common student misconceptions

- *“This is religious, not scientific.”* — Frame the tradition as historical observation. Pañcabhūta-like classifications of matter exist in many cultures. Indian astronomy and math are documented, peer-reviewed historical fields.
- *“The Sanskrit terms are just jargon.”* — Insist on the triple format (Sanskrit / IAST / meaning) once, then use the Sanskrit alone. The terms are precise; the discipline of using them properly is itself a learning outcome.
- *“If it’s traditional, it must work.”* — Distinguish empirical claims (testable) from ethical/aesthetic frames (which need separate justification). Encourage students to ask “how would I test this?”.

Differentiation strategies

- **Extension for advanced students:** offer one optional “deeper read” per day — usually a primary source or modern paper.
- **Support for students needing scaffolds:** more pair-work, allow oral responses, provide partial outlines for writing prompts.
- **Multilingual classrooms:** Sanskrit terms are universal; vernacular equivalents (Hindi, Tamil, Telugu, Bengali, Marathi etc.) should be welcomed.

Safety

- Any activity involving food, plants, fire, or outdoor work needs explicit safety briefing.
- See the relevant day file for activity-specific safety notes.

Honest framing

- Where modern science clearly extends or revises traditional claims, say so.
- Where the tradition’s framing is best understood as ethical/aesthetic, present it as such (not as competing science).
- Where attribution is debated (e.g., the “Vedic” status of Tirthaji’s mathematics), present the debate honestly.

End-of-module reflection (for teachers)

- Which concepts landed well? Which didn’t?
- Which students grew most? Which need a different approach next module?
- Add notes to `teacher-notes.md` for the next teacher who uses this pack.

Sources

Sources — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Primary and secondary sources

- Bharati Krishna Tirthaji, *Vedic Mathematics* (1965, posthumous; manuscript 1911–1918)
- S. G. Dani, “Myths and reality: On Vedic mathematics,” *Frontline* (1993) — critical historical perspective
- Kenneth R. Williams, *Vedic Mathematics — Teacher’s Manual* (2002)
- NCERT Mathematics Class VI, ch. 2 (Whole numbers)

A note on citation discipline

Per the curriculum’s style guide, citations must be verifiable. If a verse number, page, or attribution cannot be confirmed, the source is paraphrased without invented attribution. Where direct quotes appear in Senior-band day files, the source is identified in IAST + edition.

For deeper reading

- The shared glossary at [curriculum/_shared/glossary.md](#) contains the IAST + meaning for every Sanskrit term used.
- Other modules in this curriculum that connect: see the module README's "Prerequisites" section.

Day 1 — The Hook — What Is the Trick?

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 23×11 , 35×11 mentally using the split-and-sum method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)

Primary source (today's reading)

From *Bharati Krishna Tirthaji* — see [sources.md](#).

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What's one thing you remember about yesterday's lesson?*”
- Hook: introduce a phenomenon or question that motivates today's concept.

Core (20 min)

1. **Compute 23×11 silently in your head.** Pause. Show of hands for who got 253. Then write $23 \times 11 = 253$ on the board.
2. **Reveal the trick.** Split the digits of 23 (gap in the middle): **2 _ 3**. Add the digits in the middle: $2 + 3 = 5$. Fill the gap: **2 5 3** = 253. Done.
3. **Try together.** $35 \times 11 = 3_ (3+5) _ 5 = 385$. $24 \times 11 = 264$. $71 \times 11 = 781$. $52 \times 11 = 572$. $80 \times 11 = 880$. Pace: 4 problems in 2 minutes.

4. **Tirthaji's framing.** This method appears as an application of the sūtra *Ekādhikena Pūrveṇa* (“by one more than the previous”) in Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965). Honest historical note: the manuscript was written 1911–1918; some scholars (Dani 1993) consider it a 20th-century mental-arithmetic system rather than strictly ancient mathematics.
5. **Independent practice (5 min).** Worksheet: 20 two-digit $\times$ 11 problems, no carry case. Pairs compete.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Hook — What Is the Trick?.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — The Carry Case

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 87×11 , 65×11 correctly using the carry rule.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: sūtra; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Try 87×11 .** Apply the rule: $8 _ (8+7) _ 7 = 8 _ 15 _ 7$. But 15 doesn’t fit in one slot. So *carry*: keep the 5 in the middle, add the 1 to the 8 $\rightarrow 9 \ 5 \ 7 = \mathbf{957}$.
2. **General rule extension.** When the middle sum ≥ 10 :
 - Write the units of the sum in the middle.
 - Carry the tens digit to the left, adding it to the existing left digit.
3. **Practice progression.**
 - $65 \times 11 = 6 _ (11) _ 5 \rightarrow 715$
 - $49 \times 11 = 4 _ (13) _ 9 \rightarrow 539$
 - $99 \times 11 = 9 _ (18) _ 9 \rightarrow 1089$
 - $84 \times 11 = 9 _ (12) _ 4 \rightarrow 924$
4. **Cross-check with standard long multiplication.** Have students verify two of the answers using the traditional algorithm. “*Same answer? Good. The Vedic method is just a different procedure for the same arithmetic.*”
5. **Limit recognition.** “*When might this be slower than long multiplication?*” For numbers with three or more digits where carries cascade, the saving shrinks. Tomorrow we extend to 3-digit and meet the algebraic proof.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces The Carry Case.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 3 — Why It Works — The Algebra

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: derive $(10a+b) \times 11 = 100a + 10(a+b) + b$ and explain.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for ×12, ×13 extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Write any 2-digit number as $10a + b$.** Examples: $23 = 10 \cdot 2 + 3$ ($a=2$, $b=3$); $87 = 10 \cdot 8 + 7$.

2. **Distribute multiplication.**

$$\begin{aligned}(10a + b) \times 11 &= (10a + b)(10 + 1) \\ &= (10a + b) \cdot 10 + (10a + b) \cdot 1 \\ &= 100a + 10b + 10a + b \\ &= 100a + 10(a + b) + b\end{aligned}$$

3. **Read the final line in plain language.**

- $100a \rightarrow$ hundreds digit is a
- $10(a+b) \rightarrow$ tens digit is $a+b$ (the digit sum!)
- $+b \rightarrow$ units digit is b

This IS the trick. The algebra spells out the procedure.

4. **Verify on 23.** $100 \cdot 2 + 10 \cdot (2+3) + 3 = 200 + 50 + 3 = 253$. ✓
5. **Connection to the carry case.** When $a+b \geq 10$, $10(a+b)$ becomes $10(a+b) = 100 + 10 \cdot (a+b-10)$, pushing the overflow into the hundreds place. The algebra automatically handles the carry.
6. **Independent task.** Each student writes the proof in their own words for a homework explainer to a younger student.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why It Works — The Algebra.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — 3-Digit $\times$ 11

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute $234 \times 11 = 2_5_7_4$ using the extended split method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: Ekādhikena Pūrveṇa; lit. ““by one more than the previous” — sūtra 1 in Tirthaji”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Generalise to 3-digit.** For abc ($= 100a + 10b + c$) $\times 11$: $(100a + 10b + c)(10 + 1) = 1000a + 100b + 10c + 100a + 10b + c = 1000a + 100(a+b) + 10(b+c) + c$
2. **Procedure.** Split abc with two gaps: **a _ b _ c**. Fill gaps with adjacent sums: $a+b$, then $b+c$. Result: **a (a+b) (b+c) c**. Carry where needed.
3. **Worked examples.**
 - $234 \times 11 = 2_ (2+3)_ (3+4)_ 4 = 2574$
 - $567 \times 11 = 5_ (5+6)_ (6+7)_ 7 = 6237$ (carries: $11 \rightarrow$ carry 1 to left; $13 \rightarrow$ carry 1 to next left)
 - $729 \times 11 = 7_ (7+2)_ (2+9)_ 9 = 8019$
4. **Algebra check.** $234 = 100 \cdot 2 + 10 \cdot 3 + 4$. $\times 11 = 1000 \cdot 2 + 100 \cdot 5 + 10 \cdot 7 + 4 = 2574$. ✓
5. **Extension activity.** Each pair solves five 3-digit $\times 11$ problems, then verifies by long multiplication. Time-budget 8 minutes; share fastest correct-answer time.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces 3-Digit $\times 11$.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 5 — Speed Day

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: complete a 50-problem mental-math sprint in 5 minutes.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Setup.** 50-problem mental-math sprint. Mixed 2- and 3-digit $\times 11$. Half no-carry, half carry. Five minutes on the clock.
2. **Solo run.** Silence. Pen-on-paper. Five minutes.
3. **Pair-mark.** Swap with partner; mark in two minutes.
4. **Analyse errors.** “*Most common error?*” (Forgetting to carry; transposing digits.) “*Fastest sub-strategy?*” Discuss.

5. **Standards comparison.** Equivalent traditional long-multiplication sprint would take ~15 minutes. Time-saving on this problem class: ~3×. State the limit: this is for ×11 specifically. Tomorrow we extend to ×12 through ×19.
6. **Personal best.** Each student records their score and target for next week.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Day.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Anurūpyena — ×12 to ×19

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: extend the method to ×12 and ×13 using one-more-digit adjustments.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for ×12, ×13 extensions)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Move beyond $\times 11$.** Tirthaji’s *Anurūpyena* (“proportionately”) sūtra extends the digit-sum idea to $\times 12$ through $\times 19$. Procedure for $\times 12$:
 - Write any number N .
 - Compute: $N \times 12 = (N \times 10) + (N \times 2)$ — easy mental sum.
 - Equivalent digit-procedure: split-double-add.
2. **Worked example, 23×12 .**
 - $23 \times 10 = 230$
 - $23 \times 2 = 46$
 - Sum: 276
 - OR digit-wise: $2_3 \rightarrow 2_ (2 \cdot 1 + 3) _ (3 \cdot 1)$ wait — this gets tangled. The cleanest mental version is the $(\times 10) + (\times k)$ split.
3. **General $\times (10+k)$ rule.** $N \times (10+k) = 10N + kN$. The *Anurūpyena* framing emphasises the proportional scaling.
4. **Practice.** Compute mentally:
 - $14 \times 13 = 140 + 42 = 182$
 - $25 \times 15 = 250 + 125 = 375$
 - $32 \times 17 = 320 + 224 = 544$
5. **Honest historical note.** The $(10+k)$ decomposition isn’t unique to Tirthaji — Western mental-math systems describe the same trick. Tirthaji’s contribution is grouping it under a Sanskrit sūtra-system.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces *Anurūpyena* — $\times 12$ to $\times 19$.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 7 — Limits of the Method

Module: $\times 11$ Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: recognise when traditional long multiplication is faster than the *sūtra* method.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See *lesson-plan.md* for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: *Ekādhikena Pūrveṇa*; lit. ““by one more than the previous” — *sūtra* 1 in *Tīrthajī*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pose the question.** “*When IS the $\times 11$ (or near-base) method genuinely faster than traditional long multiplication? When is it the SAME or SLOWER?*”
2. **Categorise problem types.**
 - 2-digit $\times 11$ — much faster.
 - 3-digit $\times 11$ — somewhat faster.
 - 2-digit $\times 12$ – 19 — comparable.
 - 2-digit $\times 23$ (far from base) — long multiplication is faster.

- Long multi-digit products — long multiplication or scientific notation is best.
- 3. **Honest framing.** Tirthaji’s sūtras are not a universal speed-up. They are a *toolbox of mental shortcuts for specific patterns*. The skill is recognising when to use which tool.
- 4. **Reference.** Modern mathematicians (S.G. Dani, 1993; Kim Plofker, *Mathematics in India*, 2009) have written critically about the “Vedic” claim and pedagogically about the methods. Read Dani’s *Frontline* article excerpt aloud (one paragraph).
- 5. **Reflection.** Each student writes 3 sentences: which method-types are useful, what limits are, and where standard arithmetic remains best.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Limits of the Method.

Wrap-up (5 min)

- One-sentence exit ticket: “What’s one thing you learned today?”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 8 — Tirthaji and Historical Context

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain who tirthaji was and why scholars debate the “vedic” attribution.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sūtra* (IAST: *sūtra*; lit. “aphorism / formula”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Who was Tirthaji?** Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, Puri. Composed the manuscript that became *Vedic Mathematics* during 1911–1918. Published posthumously 1965.
2. **Tirthaji’s claim.** The 16 sūtras + 13 sub-sūtras were “reconstructed” from the *Pariśiṣṭa of the Atharva Veda*. The original manuscript with sūtras has never been independently located.
3. **Scholarly response.** S.G. Dani (Tata Institute of Fundamental Research), in *Frontline* (1993), examined the claim: the sūtras as published do NOT appear in extant Atharva Veda Pariśiṣṭa manuscripts. Most academic mathematicians treat Tirthaji’s system as a 20th-century mental-arithmetic synthesis, drawing on traditional and modern sources, NOT strictly ancient mathematics.
4. **Pedagogical value.** The methods themselves work and teach number sense effectively. The *historical claim* is separable from the *pedagogical method*. Many curricula in India today teach the methods while updating the historical claim.
5. **Reading.** Hand out a one-page excerpt from Dani’s article and Tirthaji’s preface side-by-side. Pairs discuss: “*What does each author claim? What evidence does each provide?*”

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Tirthaji and Historical Context.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — Mixed Practice

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: solve a mixed problem set using best-fit method per problem.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Anurūpyena* (IAST: Anurūpyena; lit. “proportionately” — sūtra used for $\times 12$, $\times 13$ extensions”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Mixed-method practice set.** 25 problems, 15 minutes. Each problem is either:
 - Easy $\times 11$ (no carry)
 - Tricky $\times 11$ (with carry)
 - $\times 12$ or $\times 13$ (Anurūpyena)
 - Standard 2-digit $\times$ 2-digit (use any method)
2. **Self-routing.** Students decide WHICH method to apply per problem, justified silently. Speed and accuracy both matter.
3. **Pair-mark + discuss.** Two minutes per pair to compare strategies. “*Did you both pick the same method? When did you disagree?*”

4. **Whole-class harvest.** Two students share one problem each where method choice mattered.
5. **Reflection (3 sentences).** Each student writes which method-type they're confident in, which they want more practice on, which they avoid.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Mixed Practice.

Wrap-up (5 min)

- One-sentence exit ticket: "*What's one thing you learned today?*"
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Final Assessment

Module: ×11 Multiplication — *Ekādhikena Pūrveṇa* · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: take the summative quiz and reflect on which method works best.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *Ekādhikena Pūrveṇa* (IAST: *Ekādhikena Pūrveṇa*; lit. ““by one more than the previous” — sūtra 1 in *Tīrthajī*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Method-comparison reflection (10 min).** Each student writes a one-paragraph response: “*Compare the Tirthaji $\times 11$ method with traditional long multiplication. When is each preferable? What does Tirthaji’s system teach you that standard arithmetic doesn’t?*”
2. **Honest history check (5 min).** “*In 2-3 sentences, summarise the historical debate about ‘Vedic Mathematics.’ What’s claimed? What does scholarship say?*”
3. **Summative quiz (25 min).** Mix of computation, algebraic proof, and conceptual questions. See `quizzes/summative.md`.
4. **Project share (5 min).** Each student shows the method-comparison poster they built (`assessments/project-brief.md`).
5. **Module close.** “*What’s one number-sense habit from this module that you’ll keep using?*” Each student shares one sentence.

Activity (15 min)

A scaled version of *Mental Math Sprint Cards* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Final Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Formative Quiz — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Senior)

When: mid-module (after Day 5). **Duration:** 25 minutes. **Format:** mix of short answer, multiple choice, and reflection.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of $\times 11$ Multiplication — Ekādhikena Pūrveṇa in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Why It Works — The Algebra’?
(a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *$\times 11$ Multiplication — Ekādhikena Pūrveṇa is purely a religious concept and has no scientific framing.*
5. Connect $\times 11$ Multiplication — Ekādhikena Pūrveṇa to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘Ekādhikena Pūrveṇa’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about $\times 11$ Multiplication — Ekādhikena Pūrveṇa and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Senior)

When: end of module (after Day 10). **Duration:** 45 minutes. **Total points:** 30.

Questions

Part A — Vocabulary (~5 points) A1. Define *Ekādhikena Pūrveṇa* in your own words. A2. Define *sūtra* in your own words. A3. Define *Anurūpyena* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of $\times 11$ Multiplication — Ekādhikena Pūrveṇa in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare $\times 11$ Multiplication — Ekādhikena Pūrveṇa with a concept from

another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of $\times 11$ Multiplication — Ekādhikena Pūrveṇa from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What's one question you still have?

Part E — Primary source (~5 points; Senior only) E1. Quote or paraphrase one passage from a primary source cited in this module. Explain it in your own words.

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 18/30 indicates the student has met the module's learning objectives.

Homework

Homework — $\times 11$ Multiplication — Ekādhikena Pūrveṇa (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — The Hook — What Is the Trick?** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — The Carry Case** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — 3-Digit $\times 11$** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Speed Day** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Anurūpyena — $\times 12$ to $\times 19$** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Limits of the Method** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Tirthaji and Historical Context** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Writing Prompts

Writing Prompts — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain ×11 Multiplication — Ekādhikena Pūrveṇa to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare ×11 Multiplication — Ekādhikena Pūrveṇa with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about ×11 Multiplication — Ekādhikena Pūrveṇa? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Activity 1 — Mental Math Sprint Cards

Module: ×11 Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of $\times 11$ Multiplication — Ekādhikena Pūrveṇa through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

flashcards (20–50 pre-printed $\times 11$ problems), stopwatch

What students do

In pairs: one student holds up a card, the other answers within 3 seconds using the $\times 11$ method. Swap. Track personal best time.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Method Comparison Poster

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Senior

What you will do

Choose 5 multiplication problems and solve them using both the $\times 11$ method and traditional long multiplication. Time yourself. Make a poster showing when each method is faster and why.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — ×11 Multiplication — Ekādhikena Pūrveṇa (Senior)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

Accuracy of concepts — uses Sanskrit terms correctly, gets the central idea right

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Source discipline — cites sources, distinguishes traditional framing from empirical claim

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Application — connects the module to other modules and to real-world examples

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Presentation — clear, organised, evidence-supported

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

Reflection — honest, thoughtful, identifies what was learned and what remains uncertain

- **Exceeding (4)** — Exceptional; could teach this to another student.
- **Meeting (3)** — Solid; meets the learning objective.
- **Approaching (2)** — Partial; some elements present, others missing.
- **Beginning (1)** — Significant gaps; needs re-teaching.

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.